INCH - POUND

MIL-P-43226A (MI)
28 June 1991
SUPERSEDING
MIL-P-43226 (MI)
22 April 1964

MILITARY SPECIFICATION

POLYETHER CUSHIONING MATERIAL FOAM-IN-PLACE, FLEXIBLE

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a two component liquid polyether system for producing a flexible foam cushioning material for electronic equipment.

1.2 Classification. The polyether foam shall be furnished in the following types, as specified (see 6.4).

Type		Density, Pounds/Cubic foot (lb/ft^3)
Type I	-	2.0 - 3.0
Type II	-	3.0 - 4.0
Type III	-	4.0 - 5.0

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8135

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM D 3574	-	Flexible Cellular Materials - Slab, Bonded, and Molded Urethane Foams

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials. The materials used in the manufacture of polyether foam shall consist of a polyether resin as a basic ingredient which shall be mixed with a catalyst and cured in accordance with 3.3. The flexible foam cushioning product shall conform to the requirements specified herein.

3.3 Mixing and curing. Mixing and curing shall be in accordance with the manufacturer's instructions.

3.4 Instructions. Unless otherwise specified, the supplier shall furnish one mixing and curing instruction sheet with each package. The instruction sheet shall also contain the following caution:

CAUTION: From the time mixing is begun, gases are liberated which are classed as extreme irritants. All mixing, pouring, and foaming must be done in an exhaust hood or areas having adequate ventilation. Any areas of the skin in contact with the mixed or unmixed foam shall be flushed immediately with isopropyl alcohol, or thoroughly washed with soap and water.

3.5 Physical properties. When mixed and cured as specified in 3.3, the material shall be in accordance with table I.

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TABLE I. Physical properties.

Physical properties	Type I	Type II	Type III
Density, (lbs/ft^3) +10 percent (%)	2.0 to 3.0	3.0 to 4.0	4.0 to 5.0
Tensile strength, pound force per square inch (lbf/in^2)	20	25	
Elongation, %, maximum	190	130	
Compression set, 22 hours/70°C ASTM D 3574 Test D	30	30	30
Compression deflection, lbf/in^2 maximum			
10%	.61	1.28	1.60
25%	.66	1.84	
50%	.96	2.47	

3.6 Identification. The manufacturer shall identify each resin and catalyst container as specified in 5.3.

3.7 Shelf life. Shelf life of the polyether resin and the catalyst shall be not less than 90 days, when stored in their original containers at a temperature between 18.3 and 23.8° Celsius (C).

3.8 Workmanship. Workmanship shall conform to the best commercial practice for this product. There shall be no defects which might render the material unsuitable for its intended purpose. The compound shall be homogeneous and free from lumps and coarse particles.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as

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otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When specified, (see 6.2), a sample shall be subjected to first article inspection in accordance with tests specified herein.

4.4 Quality conformance inspection. Quality conformance inspections shall consist of examination and testing of the polyether cushioning material for conformance to the requirements specified herein.

4.4.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in ASTM D 3574.

4.5 Lot formation. An inspection lot shall consist of a group of like items produced at the same place utilizing the same batches of materials, lots of components, process runs, fabrication techniques, assembly techniques, tools, equipment,

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and facilities, but shall not exceed one month's production. Upon submittal to the procuring activity for acceptance, each lot shall be accompanied by in-process inspection records and inspection results as required by the procuring activity inspector and this specification.

4.6 Examination. The examination specified herein shall be performed on inspection lots (see 4.5) of the components submitted as specified in 3.5. Sampling shall be in accordance with table II sampling plan.

TABLE II. Sampling for quality conformance.

Lot Size	Sample Size
2 to 5	All
6 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1200	19
1201 to 3200	23
3201 to 10,000	29
10,001 to 35,000	35
35,001 and over	49
In all cases: Acceptance number is ZERO Rejection number is ONE	

4.7 Classification of defects. Each sample container and contents shall be examined for defects as specified in table III.

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TABLE III. Classification of defects, containers.

Categories	Defects	Method of inspection	Sampling plan (table)
Critical	None defined		
Major			
101	Container fill	Visual	II
102	Container leakage	Visual	II
103	Marking	Visual	II
104	Workmanship	Visual	II
105	Preservation, packaging, packing and marking	Visual	II

4.8 Testing. Physical tests specified in table IV shall be performed to assure compliance with the requirements specified in table I. Materials shall be mixed and cured as specified in 3.3 in a wood or metal mold. Clean equipment shall be used in weighing and mixing each pour of foam. Mold surfaces shall be smooth and a release agent applied to facilitate removal of foamed parts.

TABLE IV. Physical tests.

Physical properties tests	ASTM test method
Density, lb/ft^3	ASTM D 3574 - A
Tensile strength, lbf/in^2 minimum	ASTM D 3574 - E
Elongation, % minimum	ASTM D 3574 - F
Compression set, 22 hours/70°C	ASTM D 3574 - D
Compression deflection, lbf/in^2 Maximum	ASTM D 3574 - C
10 %	
25 %	
50 %	

4.9 Shelf life. Manufacturers shall submit certification that the polyether resin and the catalyst will meet the requirements of this specification after storage for 90 days in their original containers at a temperature between 18.3 and 23.8°C, signed by a responsible agent of the manufacturer.

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4.10 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of MIL-STD-129.

5. PACKAGING

5.1 Unit packaging.

5.1.1 Two-part kits. Unless otherwise specified in the contract or order, the polyether resin and catalyst shall be packaged in individual containers. The ratio of the quantity contained in the polyether resin container to the quantity contained in the catalyst container shall be the same as the recommended mixing ratio of the polyether resin and catalyst.

5.2 Packing. All exterior shipping containers in the shipment shall contain the same number of the type of unit packages. Shipping containers shall be of uniform size and shall be so designed as to insure that damage is prevented during handling and shipping.

5.3 Marking.

5.3.1 Base compound containers. Each polyether resin container shall be durably and legibly marked with the following information:

(a) Title and number of specification
(b) Classification (see 1.2)
(c) Manufacturer's name and address
(d) Manufacturer's product designation
(e) Manufacturer's batch designation
(f) Manufacturer's batch designation for the catalyst that the base compound is to be used with
(g) Stock number
(h) Date of manufacture
(i) Contract number
(j) Quantity contained.

5.3.2 Catalyst container. Each catalyst container shall be durably and legibly marked with the following information:

(a) Manufacturer's name and address
(b) Manufacturer's product designation
(c) Manufacturer's batch designation

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(d) Manufacturer's batch designation for the base compound that the catalyst is to be used with
(e) Date of manufacture.

5.3.3 Exterior shipping containers. Each exterior shipping container shall be durably and legibly marked with adequate instructions to insure that damage during handling and shipping is prevented.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The polyether cushioning material covered by this specification is intended to be used as a light weight cushioning material for irregular shaped electronic equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. When first article is required (see 3.1 and 4.3)

d. Instruction sheets for mixing and curing (see 3.4)

e. Lot formation requirements (see 4.6)

f. Preservation and packaging if other than specified (see Section 5)

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive

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the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Part or identifying number (PIN). The PIN to be used for polyether cushioning material acquired to this specification is created as follows:

M -	43226 -	X
Prefix to indicate military specification	Specification number	Type of material (see 1.2)

6.5 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.6 Subject term (keyword) listing.

Packaging material
Polyurethane cellular material
Polymers

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-43226A (MI)	910628

3. DOCUMENT TITLE
POLYETHER CUSHIONING MATERIAL FOAM-IN-PLACE, FLEXIBLE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER U.S. ARMY MISSILE COMMAND	b. TELEPHONE (Include Area Code) (1) Commercial: (205) 876-6980 (2) AUTOVON: 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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